

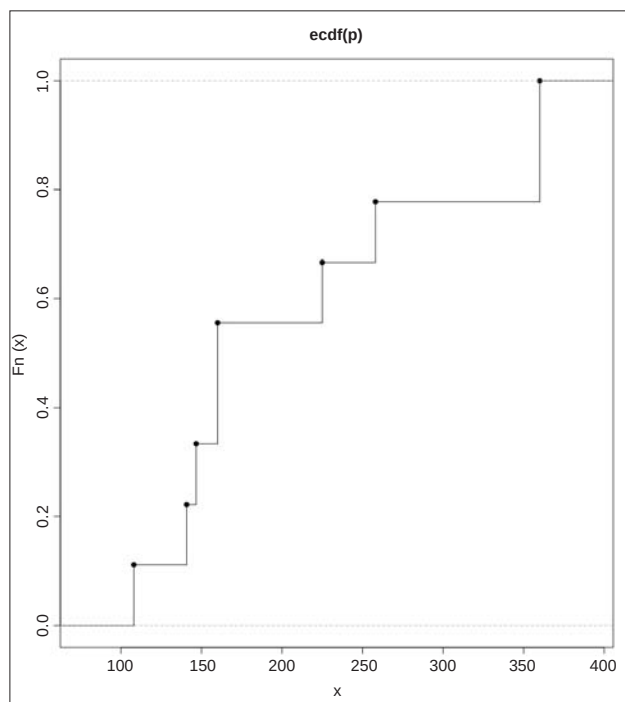


Figure 2: Empirical CDF (P)

Exact two-sample Kolmogorov-Smirnov test

data: P and Q
D = 0.37778, p-value = 0.2634
alternative hypothesis: two-sided

You can also perform a one-sample Kolmogorov-Smirnov test. The function accepts the following arguments:

Arguments	Description
x	Numeric vector of data values
y	Numeric vector or character string name of a CDF
alternative	Alternate hypothesis should be 'two-sided', 'less' or 'greater'
exact	NULL or logical value to specify if p-value should be computed
simulate.p.value	A logical value to compute p-values by Monte Carlo simulation
data	An optional data frame
subset	A subset vector of data to be used
na.action	A function to handle data that contains NAs

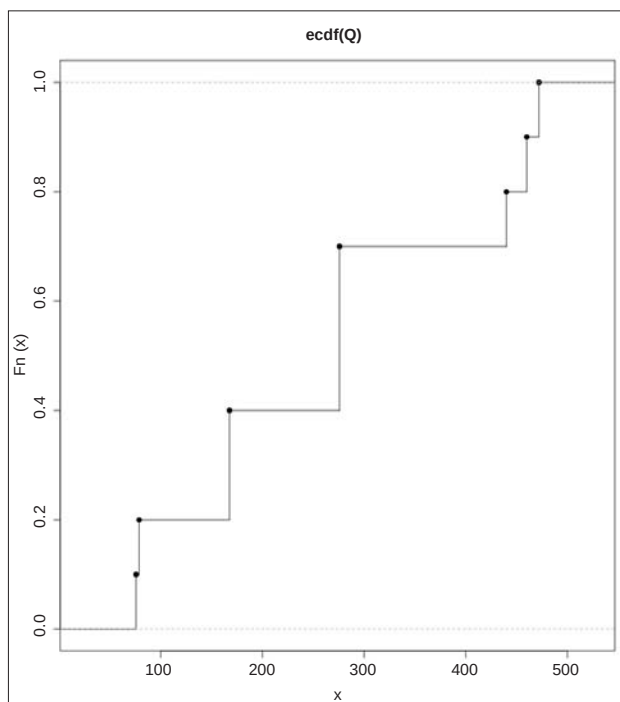


Figure 3: Empirical CDF (Q)

The alternative hypothesis argument accepts the string argument values: 'two.sided', 'less', and 'greater'. These indicate if the null hypothesis of the distribution of 'x' is equal to, less than, or greater than the distribution of 'y'. Any missing values are ignored from both the samples.

Chi-Square test

The Chi-Square test is a statistical hypothesis test to determine the correlation between two categorical variables. We can determine if the cylinder and horse power variables from the *mtcars* data set are significantly correlated.

```
> data <- table(mtcars$ cyl, mtcars$hp)
> print(data)
```

```
      52 62 65 66 91 93 95 97 105 109 110 113 123 150 175
180 205 215 230 245 264
  4  1  1  1  2  1  1  1  0  1  0  1  0  0  0
0  0  0  0  0  0  0
  6  0  0  0  0  0  0  0  1  0  3  0  2  0  1
0  0  0  0  0  0
  8  0  0  0  0  0  0  0  0  0  0  0  0  2  2
3  1  1  1  2  1
```

The *chisq.test()* function can be performed on the data variables, as shown below: